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(54) DISK DEVICE

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(57) **ABSTRACT**

A disk device includes a magnetic disk, a base, a film, an inner cover, and an outer cover. The base is provided with an internal space, and has a first inner surface surrounding the magnetic disk, a supporting surface connected to an end of the first inner surface in a first direction, a second inner surface further away from a rotation axis of the magnetic disk than the first inner surface and connected to the supporting surface, and an end face connected to an end of the second inner surface in the first direction. The film covers a covered region in the second inner surface and is spaced apart from the end face and an exposed region of the second inner surface. The inner cover is supported by the supporting surface and surrounded by the second inner surface. The outer cover is coupled to a portion of the base.

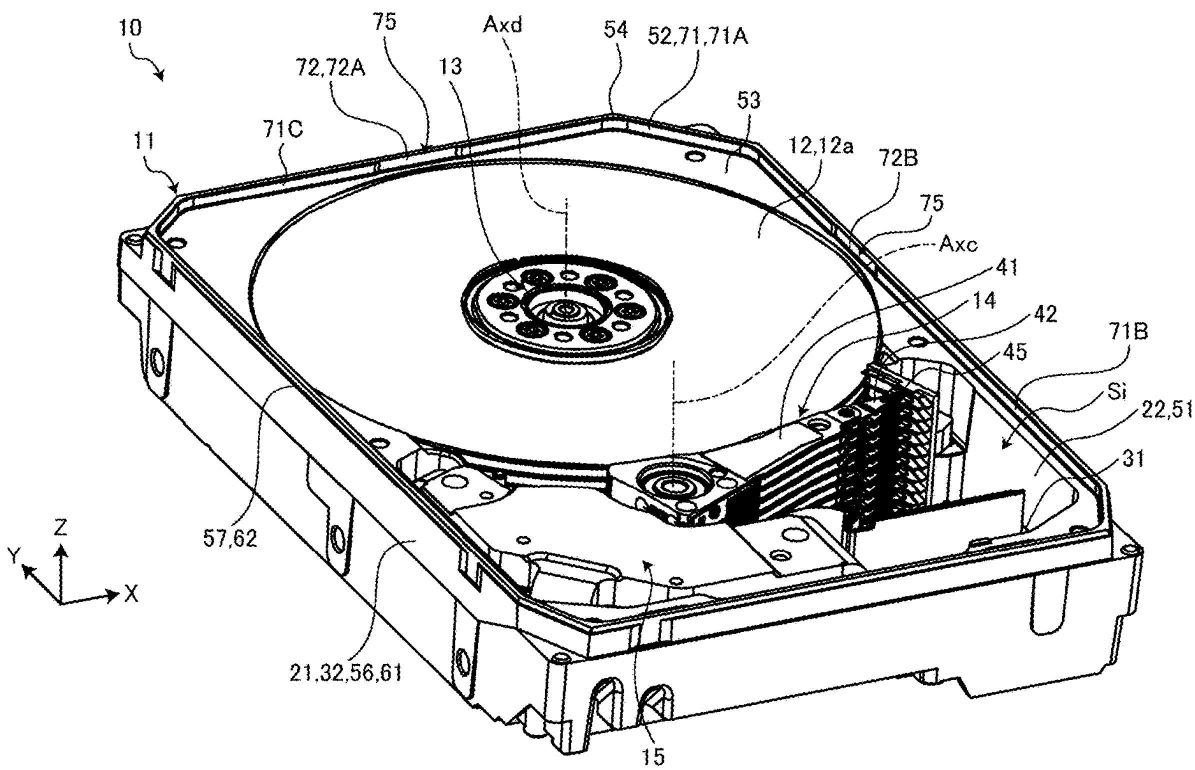
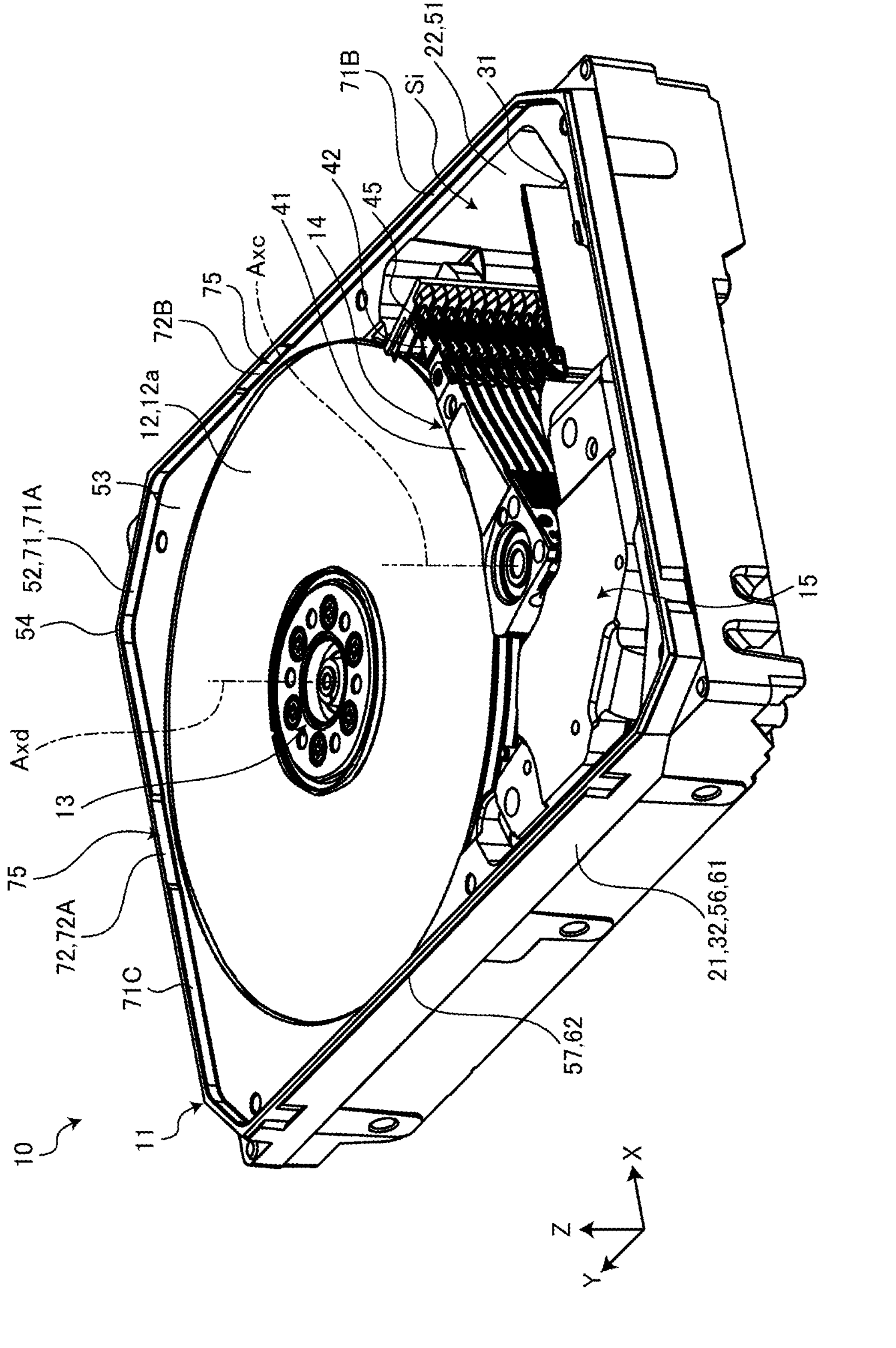
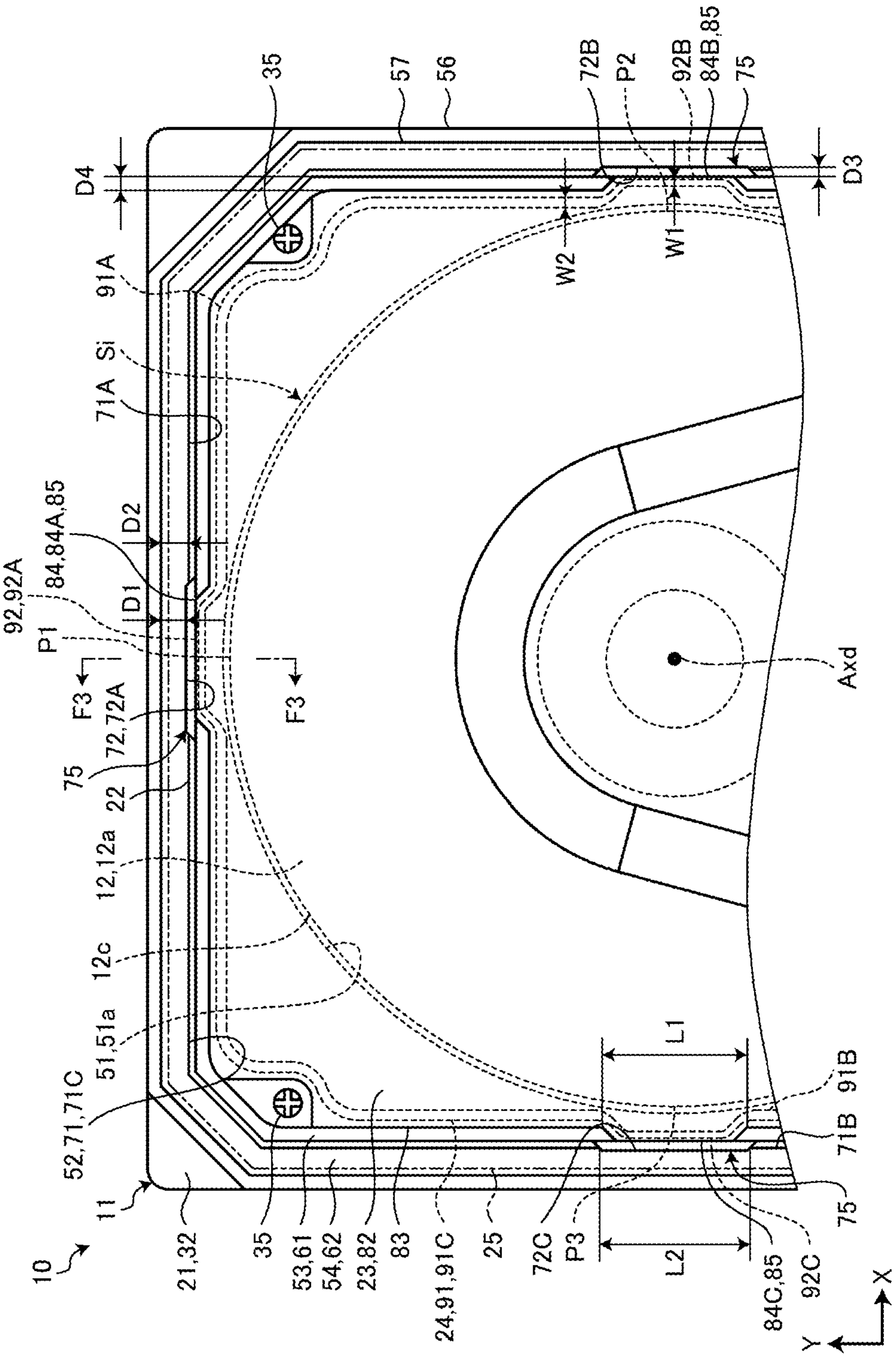


FIG. 1



22,51
71B
Si
31
42
45
41
14
Axc
75
72B
12,12a
53
52,71,71A
54
Axd
13
75
72,72A
71C
11
10
15
21,32,56,61
57,62
X
Z
Y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53,61
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72C
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92C
75
71B
91B
52,71,71C
51,51a
12c
12,12a
22
75
72,72A
F3
P1
92,92A
84,84A,85
D1
D2
F3
71A
Si
91A
35
D4
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P2
92B
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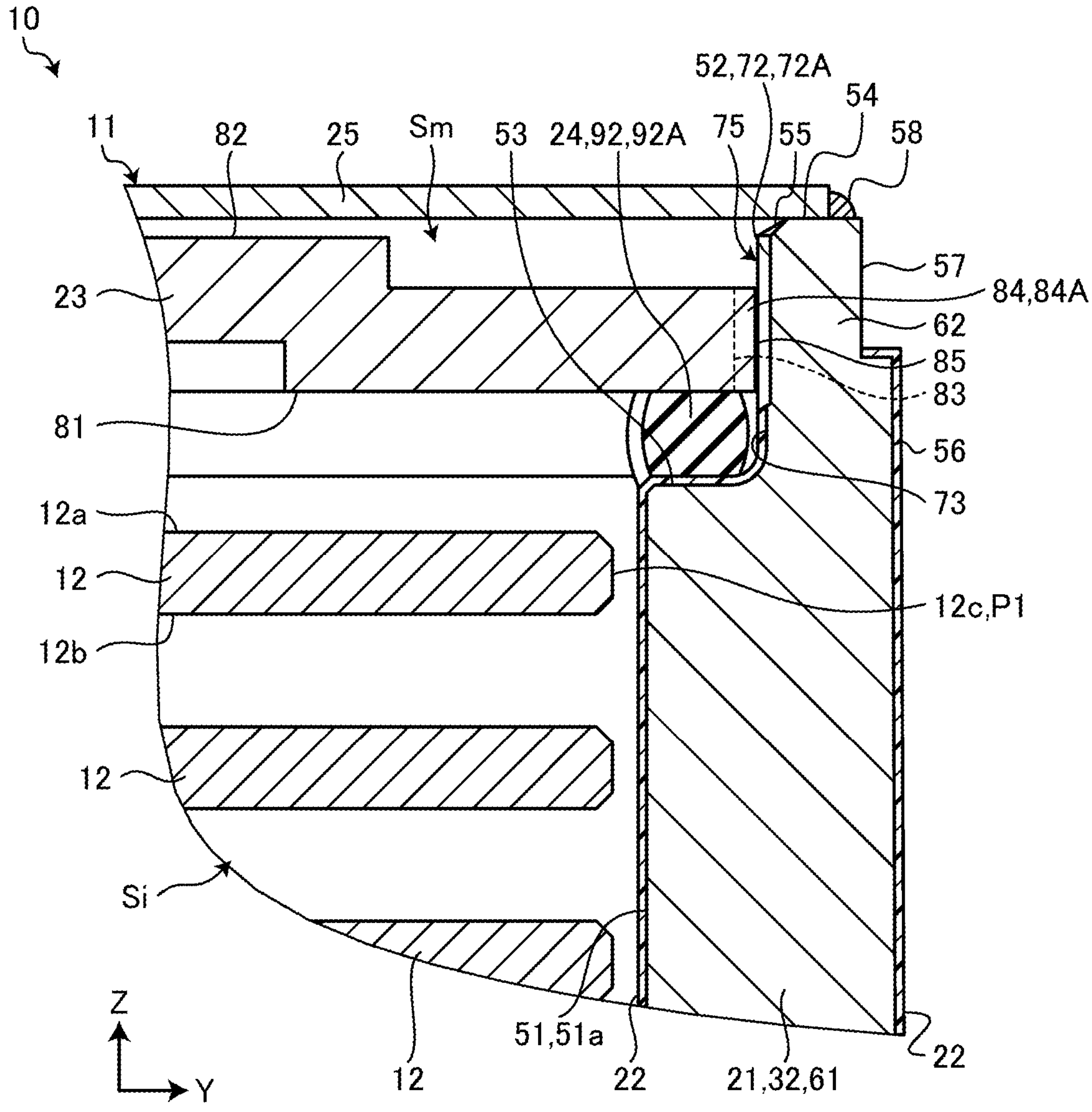
D3
W2
W1
Axd
X
Y

FIG. 2

FIG. 3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56
73
12c,P1
23
81
12a
12
12b
12
Si
Z
Y
12
51,51a
22
21,32,61
22

DISK DEVICE

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-031333, filed Mar. 1, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a disk device.

BACKGROUND

[0003] A disk device such as a hard disk drive (HDD) includes a magnetic disk and a housing that accommodates the magnetic disk. The housing includes, for example, a base, an inner cover that closes an internal space of the base, and an outer cover that covers the inner cover and is welded to an end face of the base.

[0004] The surface of the base is generally protected by an electrodeposition-coated film. On the other hand, in the base, the film may be removed in the vicinity of the outer cover, for example, by cutting, to prevent the film from melting during welding. When the width of the end face of the base is reduced by cutting, there is a concern that the strength of welding between the end face and the outer cover may be reduced.

DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a perspective view showing an HDD according to one embodiment.

[0006] FIG. **2** is a plan view schematically showing a portion of the HDD according to the embodiment.

[0007] FIG. **3** is a cross-sectional view showing a portion of the HDD according to the embodiment along line F**3**-F**3** in FIG. **2**.

DETAILED DESCRIPTION

[0008] Embodiments provide a disk device that can improve the strength of coupling between an end face of a base and an outer cover.

[0009] In general, according to one embodiment, a disk device includes a magnetic disk, a base, a film, an inner cover, and an outer cover. The magnetic disk is rotatable around a rotation axis. The base is provided with an internal space in which the magnetic disk is disposed, and has a first inner surface surrounding the magnetic disk, a supporting surface connected to an end of the first inner surface in a first direction parallel to the rotation axis, a second inner surface further away from the rotation axis than the first inner surface and connected to the supporting surface, and an end face connected to an end of the second inner surface in the first direction. The film covers at least a portion of the first inner surface, at least a portion of the supporting surface, and a covered region of the second inner surface, and is spaced apart from the end face and an exposed region of the second inner surface. The inner cover is supported by the supporting surface, is surrounded by the second inner surface, and covers the internal space. The outer cover is coupled to at least a portion of the base and covers the inner cover.

[0010] One embodiment will be described below with reference to FIGS. **1** to **3**. In this specification, elements of the embodiment and a description of the elements may be expressed in various ways. The elements and the description thereof are examples and are not limited by the expressions in this specification. The elements may also be specified by names different from those in this specification. The elements may also be described using expressions different from the expressions in this specification.

[0011] In the following description, "suppress" is defined as, for example, preventing the occurrence of an event, effect, or influence, or reducing the degree of an event, effect, or influence.

[0012] FIG. **1** is a perspective view showing a hard disk drive (HDD) **10** according to the present embodiment. FIG. **2** is a plan view schematically showing a portion of the HDD **10** according to the present embodiment. FIG. **3** is a cross-sectional view showing a portion of the HDD **10** according to the present embodiment along line F**3**-F**3** in FIG. **2**.

[0013] As shown in the drawings, an X-axis, a Y-axis, and a Z-axis are defined in this specification for convenience of description. The X-axis, the Y-axis, and the Z-axis are orthogonal to each other. The X-axis is provided along the width of the HDD **10**. The Y-axis is provided along the length of the HDD **10**. The Z-axis is provided along the thickness of the HDD **10**.

[0014] Further, in this specification, an X direction, a Y direction, and a Z direction are defined. The X direction is a direction along the X-axis, and includes a +X direction indicated by an arrow on the X-axis and a −X direction which is a direction opposite to the arrow on the X-axis. The Y direction is a direction along the Y-axis, and includes a +Y direction indicated by an arrow on the Y-axis and a −Y direction which is a direction opposite to the arrow on the Y-axis. The Z direction is a direction along the Z-axis and includes a +Z direction indicated by an arrow on the Z-axis and a −Z direction which is a direction opposite to the arrow on the Z-axis.

[0015] As shown in FIG. **1**, the HDD **10** includes a housing **11**, a plurality of magnetic disks **12**, a spindle motor **13**, a head stack assembly (HSA) **14**, and a voice coil motor (VCM) **15**. The magnetic disk **12** may also be referred to as a disk or a platter.

[0016] As shown in FIG. **3**, the housing **11** includes a base **21**, a paint film **22**, an inner cover **23**, a gasket **24**, and an outer cover **25**. The housing **11** is not limited to this example. The paint film **22** is an example of a film. For convenience of description, FIG. **1** shows the housing **11** with the inner cover **23**, the gasket **24**, and the outer cover **25** omitted. For convenience of description, FIG. **2** hypothetically shows the outer cover **25** indicated by an alternating two dots-dashed line.

[0017] Each of the base **21**, the inner cover **23**, and the outer cover **25** is made of a metal material such as an aluminum alloy. The base **21**, the inner cover **23**, and the outer cover **25** may be made of different materials in other embodiments.

[0018] The base **21** is formed in the form of a substantially rectangular parallelepiped box that is open in the +Z direction. That is, the base **21** is provided with an internal space Si. The internal space Si communicates with the outside of the base **21** through the end of the base **21** in the +Z direction. A plurality of magnetic disks **12**, the spindle motor **13**, the HSA **14**, and the VCM **15** are disposed in the internal space Si.

[0019] As shown in FIG. **1**, the base **21** extends in the Y direction (+Y direction and −Y direction). For this reason, the Y direction is referred to as the longitudinal direction of the base **21** and the housing **11**. Further, the X direction is referred to as the lateral direction of the base **21** and the housing **11**.

[0020] The base **21** includes a bottom wall **31** and a side wall **32**. The bottom wall **31** is formed in the form of a substantially rectangular plate shape disposed to be substantially orthogonal to the Z direction. The side wall **32** protrudes from the edge of the bottom wall **31** substantially in the +Z direction and is formed in a substantially rectangular frame shape.

[0021] The paint film **22** is made of, for example, an epoxy resin. The paint film **22** is not limited to this example. The paint film **22** is provided on the surface of the base **21** by, for example, electrodeposition coating. The paint film **22** has higher corrosion resistance and rust prevention than those of the base **21** and protects the base **21**.

[0022] As shown in FIG. **2**, the inner cover **23** is attached to the side wall **32** with fastening members **35** such as screws. Thereby, the inner cover **23** covers the internal space Si of the base **21**. As shown in FIG. **3**, the gasket **24** is interposed between the side wall **32** and the inner cover **23**, and seals a gap between the side wall **32** and the inner cover **23**. The outer cover **25** covers the inner cover **23** and is attached to the side wall **32**, for example, by welding. The outer cover **25** may be coupled to other portions of the base **21**. That is, the outer cover **25** is coupled to at least a portion of the base **21**.

[0023] For example, the inner cover **23** and the outer cover **25** are provided with vent holes. After the magnetic disk **12**, the spindle motor **13**, the HSA **14**, and the VCM **15** are disposed in the internal space Si, and the inner cover **23** and outer cover **25** are attached to the base **21**, air in the housing **11** is removed through the vent holes. Furthermore, the internal space Si is filled with a gas different from air. The vent hole of the outer cover **25** is airtightly sealed, for example, with a seal.

[0024] The gas filled in the internal space Si is, for example, a low-density gas whose density is lower than that of air, an inert gas with low reactivity, or the like. For example, the internal space Si is filled with helium. The internal space Si may be filled with other fluids.

[0025] The plurality of magnetic disks **12** are arranged with gaps in the Z direction. The plurality of magnetic disks **12** are formed in a disk shape disposed substantially orthogonal to the Z direction. The diameter of each of the plurality of magnetic disks **12** is, for example, 96 mm or more. The diameter of the magnetic disk **12** is not limited to this example.

[0026] Each of the plurality of magnetic disks **12** has two flat surfaces **12***a* and **12***b* and an outer peripheral surface **12***c*. One flat surface **12***a* faces substantially in the +Z direction. The other flat surface **12***b* is located on a side opposite to the flat surface **12***a* and faces substantially in the −Z direction. The outer peripheral surface **12***c* is a substantially cylindrical curved surface extending substantially in the Z direction between the outer edge of the flat surface **12***a* and the outer edge of the flat surface **12***b*. A magnetic recording layer is provided on the flat surfaces **12***a* and **12***b*.

[0027] The spindle motor **13** in FIG. **1** supports the plurality of magnetic disks **12**. The plurality of magnetic disks **12** are held on a hub of the spindle motor **13**, for example, by a clamp spring. The spindle motor **13** rotates the plurality of magnetic disks **12** integrally around a central axis Axd. The central axis Axd is an example of a rotation axis.

[0028] The central axis Axd is a virtual axis extending substantially in the Z direction. The central axis Axd is, for example, the central axis of the magnetic disk **12** and the spindle motor **13**. The central axis Axd is not limited to this example.

[0029] The HSA **14** includes a carriage **41** and a plurality of head gimbal assemblies (HGA) **42**. The HSA **14** is not limited to this example. The carriage **41** is attached to the base **21** to be rotatable around a central axis Axc. The central axis Axc is a virtual axis extending substantially in the Z direction. That is, the central axis Axc of the carriage **41** and the central axis Axd of the magnetic disk **12** extend substantially parallel to each other. The central axis Axc is spaced apart from the central axis Axd.

[0030] The plurality of HGAs **42** are attached to an arm of the carriage **41** and rotate integrally with the carriage **41** around the central axis Axc. The plurality of HGAs **42** are arranged with gaps in the Z direction. Each of the plurality of HGAs **42** includes a magnetic head **45**. The magnetic head **45** may also be referred to as a slider.

[0031] The magnetic head **45** records and reproduces information for a corresponding magnetic disk among the plurality of magnetic disks **12**. In other words, the magnetic head **45** reads and writes information from and to the magnetic disk **12**. The carriage **41** moves the magnetic head **45** with respect to the corresponding magnetic disk **12** by rotating around the central axis Axc.

[0032] The VCM **15** moves the magnetic head **45** to a desired position along the magnetic disk **12** by rotating the carriage **41** around the central axis Axc. The VCM **15** includes a voice coil, a pair of yokes, and magnets provided on the yokes. The voice coil is held by the carriage **41**.

[0033] As shown in FIG. **3**, the side wall **32** has a first inner surface **51**, a second inner surface **52**, a supporting surface **53**, an end face **54**, an inclined surface **55**, a first outer surface **56**, and a second outer surface **57**. The side wall **32** is not limited to this example.

[0034] Each of the first inner surface **51**, the second inner surface **52**, the supporting surface **53**, the end face **54**, the inclined surface **55**, the first outer surface **56**, and the second outer surface **57** is formed in a frame shape surrounding the central axes Axc and Axd. The first inner surface **51** and the second inner surface **52** face in a direction substantially orthogonal to the central axis Axd. At least portions of the first inner surface **51** and the second inner surface **52** may face in other directions.

[0035] The first inner surface **51** extends from the bottom wall **31** substantially in the +Z direction. The first inner surface **51** surrounds the plurality of magnetic disks **12**, at least a portion of the spindle motor **13**, at least a portion of the HSA **14**, and at least a portion of the VCM **15**. That is, in the Z direction, the plurality of magnetic disks **12** are disposed between the end of the first inner surface **51** in the +Z direction and the end of the first inner surface **51** in the −Z direction. A portion of the magnetic disk **12** may protrude in the +Z direction beyond the end of the first inner surface **51** in the +Z direction in other embodiments.

[0036] The first inner surface **51** has a curved surface **51***a*. The curved surface **51***a* is a substantially cylindrical curved surface that extends along the outer peripheral surface **12***c* of the magnetic disk **12**. In other words, the curved surface **51***a* of the first inner surface **51** and the outer peripheral surface **12***c* of the magnetic disk **12** are disposed concentrically (coaxially).

[0037] The curved surface **51***a* functions as a so-called shroud. For this reason, the curved surface **51***a* regulates a flow of helium in the internal space Si in the vicinity of the magnetic disk **12**, thereby preventing a turbulent flow of helium from being generated.

[0038] The second inner surface **52** is further away from the bottom wall **31** than the first inner surface **51**. Furthermore, the second inner surface **52** is further away from the central axes Axc and Axd than the first inner surface **51** is. For this reason, the second inner surface **52** surrounds the first inner surface **51** in a projection plane viewed in the Z direction (+Z direction or −Z direction) as shown in FIG. **2**.

[0039] As shown in FIG. **3**, the supporting surface **53** and the end face **54** face substantially in the +Z direction. The supporting surface **53** and the end face **54** may be provided with irregularities. That is, at least portions of the supporting surface **53** and the end face **54** may face in other directions.

[0040] The supporting surface **53** is connected to the end of the first inner surface **51** in the +Z direction and the end of the second inner surface **52** in the −Z direction. In other words, the supporting surface **53** is provided between the first inner surface **51** and the second inner surface **52**. The supporting surface **53** supports the inner cover **23** via the gasket **24**. That is, the gasket **24** is interposed between the supporting surface **53** and the inner cover **23**.

[0041] The second inner surface **52** surrounds the inner cover **23** and the gasket **24** supported by the supporting surface **53**. That is, the inner cover **23** and the gasket **24** are disposed between the end of the second inner surface **52** in the +Z direction and the end of the second inner surface **52** in the −Z direction. In other embodiments, a portion of the inner cover **23** may protrude in the +Z direction beyond the end of the second inner surface **52** in the +Z direction, and a portion of the gasket **24** may protrude in the −Z direction beyond the end of the second inner surface **52** in the −Z direction.

[0042] The first inner surface **51** is provided in the internal space Si. In other words, the bottom wall **31** and the first inner surface **51** form (define, partition) the internal space Si. The second inner surface **52** is provided in an intermediate space Sm between the inner cover **23** and the outer cover **25**. The supporting surface **53** may be partially provided in the internal space Si, or may be partially provided in the intermediate space Sm.

[0043] The end face **54** is connected to the end of the second inner surface **52** in the +Z direction. The end face **54** is provided at the end of the side wall **32** in the +Z direction. The side wall **32** may have another portion located at the end of the side wall **32** in the +Z direction.

[0044] The outer cover **25** is coupled to the end face **54**, for example, by welding. For this purpose, a bead **58** is provided on the end face **54**. The bead **58** extends along the end face **54**. The outer cover **25** may be coupled to the end face **54** by other methods.

[0045] The inclined surface **55** is provided between the second inner surface **52** and the end face **54**. That is, the end face **54** is connected to the end of the second inner surface **52** in the +Z direction via the inclined surface **55**. The end face **54** may be directly connected to the second inner surface **52** in other embodiments.

[0046] The inclined surface **55** extends obliquely between the second inner surface **52** and the end face **54** with respect to the second inner surface **52** and the end face **54**. The inclined surface **55** is formed by, for example, a chamfer. The inclined surface **55** is not limited to this example.

[0047] The first outer surface **56** is located on a side opposite to the first inner surface **51** and the second inner surface **52**. The second outer surface **57** is located on a side opposite to the second inner surface **52**. The end of the second outer surface **57** in the +Z direction is connected to the end face **54**. The end of the second outer surface **57** in the −Z direction is connected to the first outer surface **56**.

[0048] The side wall **32** includes a wall **61** and a rib **62**. The wall **61** has the first inner surface **51**, the supporting surface **53**, and a portion of the first outer surface **56**. That is, the wall **61** protrudes from the edge of the bottom wall **31** substantially in the +Z direction. The supporting surface **53** is an end face of the wall **61** in the +Z direction. The rib **62** has the second inner surface **52**, the end face **54**, a portion of the first outer surface **56**, and the second outer surface **57**. The rib **62** protrudes from the supporting surface **53** of the wall **61** substantially in the +Z direction. The outer cover **25** is welded to the rib **62**.

[0049] As shown in FIG. **2**, the second inner surface **52** includes a covered region **71** and an exposed region **72**. As shown in FIG. **3**, the second inner surface **52** further includes a lower region **73**. The second inner surface **52** is not limited to this example.

[0050] As shown in FIG. **2**, the covered region **71** and the exposed region **72** are adjacent to each other around the central axis Axd. In other words, the covered region **71** and the exposed region **72** are disposed around the central axis Axd. The covered region **71** and the exposed region **72** are at least partially disposed at substantially the same position (height) in the Z direction.

[0051] In the present embodiment, the covered region **71** has three covered regions **71**A, **71**B, and **71**C. The covered region **71** is not limited to this example. The three covered regions **71**A, **71**B, and **71**C are spaced apart from each other around the central axis Axd.

[0052] In the present embodiment, the exposed region **72** has three exposed regions **72**A, **72**B, and **72**C. The exposed region **72** is not limited to this example. The three exposed regions **72**A, **72**B, and **72**C are spaced apart from each other around the central axis Axd.

[0053] The three covered regions **71**A, **71**B, and **71**C and the three exposed regions **72**A, **72**B, and **72**C are disposed alternately around the central axis Axd. The exposed region **72**A is located between the two covered regions **71**A and **71**C. The exposed region **72**B is located between the two covered regions **71**A and **71**B. The exposed region **72**C is located between the two covered regions **71**B and **71**C.

[0054] In the projection plane viewed in the Z direction as shown in FIG. **2**, the exposed region **72**A faces an end (apex) P1 of the magnetic disk **12** in the +Y direction.

[0055] The central axis Axd of the magnetic disk **12** is spaced apart in the +Y direction from the center of the base **21** in the Y direction. That is, the magnetic disk **12** is spaced apart from the end of the base **21** in the −Y direction and closer to the end of the base **21** in the +Y direction. On the other hand, the central axis Axc of the carriage **41** is spaced apart from the end of the base **21** in the +Y direction and closer to the end of the base **21** in the −Y direction.

[0056] In the projection plane viewed in the Z direction, the exposed region **72**B faces an end P**2** of the magnetic disk **12** in the +X direction.

[0057] In the projection plane viewed in the Z direction, the exposed region **72**C faces an end P**3** of the magnetic disk **12** in the −X direction. The two exposed regions **72**B and **72**C face each other.

[0058] The exposed region **72** is closer to the magnetic disk **12** than the covered region **71**. For example, the exposed region **72**A is a portion of the second inner surface **52** which is closest to the end P**1** of the magnetic disk **12**. The exposed region **72**B is a portion of the second inner surface **52** which is closest to the end P**2** of the magnetic disk **12**. The exposed region **72**C is a portion of the second inner surface **52** which is closest to the end P**3** of the magnetic disk **12**.

[0059] A distance between the exposed region **72**A and the magnetic disk **12**, a distance between the exposed region **72**B and the magnetic disk **12**, and a distance between the exposed region **72**C and the magnetic disk **12** are substantially equal to each other. The distances may be different from each other in other embodiments.

[0060] As shown in FIG. **3**, the lower region **73** is provided between the covered region **71** and the supporting surface **53** and between the exposed region **72** and the supporting surface **53**. For this reason, the covered region **71** and the exposed region **72** are spaced apart from the supporting surface **53** substantially in the +Z direction. The lower region **73** is provided over the entire circumference around the central axis Axd. The lower region **73** is not limited to this example.

[0061] The paint film **22** covers at least a portion of the first inner surface **51**, at least a portion of the supporting surface **53**, the first outer surface **56**, the covered region **71** (covered regions **71**A, **71**B, and **71**C), and at least a portion of the lower region **73**. In other words, the paint film **22** adheres to the first inner surface **51**, the supporting surface **53**, the first outer surface **56**, the covered region **71**, and the lower region **73**.

[0062] The paint film **22** does not cover the end face **54**, the second outer surface **57**, and the exposed region **72** (exposed regions **72**A, **72**B, and **72**C), so as to expose them. In other words, the paint film **22** is spaced apart from the end face **54**, the second outer surface **57**, and the exposed region **72**. The end face **54**, the second outer surface **57**, and the exposed region **72** may be covered with another component such as the outer cover **25**.

[0063] For example, the exposed region **72** is formed on the second inner surface **52** by removing the paint film **22** provided on the second inner surface **52** by cutting. A portion of the second inner surface **52** is cut to become the exposed region **72** that is recessed from the other portions, which are the covered region **71** and the lower region **73**. For this reason, the exposed region **72** forms a concave portion **75** that opens to the end face **54**, the covered region **71**, and the lower region **73**. The covered region **71**, the exposed region **72**, and the lower region **73** may form the same plane.

[0064] As shown in FIG. **2**, the width of the portion of the rib **62** where the exposed region **72** is provided is smaller than the width of a portion of the rib **62** where the covered region **71** is provided. That is, a distance D**1** between the exposed region **72** and the second outer surface **57** is smaller than a distance D**2** between the covered region **71** and the second outer surface **57**. The distances D**1** and D**2** are not limited to this example.

[0065] As shown in FIG. **3**, the length of the inclined surface **55** between the exposed region **72** and the end face **54** is smaller than the length of the inclined surface **55** between the covered region **71** and the end face **54**. The length of the inclined surface **55** is not limited to this example.

[0066] In the Z direction, the inner cover **23** is disposed between the end of the exposed region **72** in the +Z direction and the end of the exposed region **72** in the −Z direction. For this reason, the covered region **71** and the exposed region **72** face the inner cover **23**. Furthermore, the inner cover **23** is further away from the supporting surface **53** than the end of the exposed region **72** in the −Z direction.

[0067] The lower region **73** faces the gasket **24**. In the Z direction, the lower region **73** is spaced apart from the inner cover **23** in the −Z direction. For this reason, the lower region **73** does not face the inner cover **23**. The lower region **73** is not limited to this example.

[0068] The inner cover **23** has an inner surface **81** and an outer surface **82**. As shown in FIG. **2**, the inner cover **23** further has a side surface **83** and a convex portion **84**. The inner cover **23** is not limited to this example. The inner cover **23** may not be provided with the convex portion **84** in other embodiments.

[0069] As shown in FIG. **3**, the inner surface **81** faces the bottom wall **31** and the supporting surface **53**. The gasket **24** is interposed between the supporting surface **53** and the inner surface **81**. The inner surface **81** faces the internal space Si. The outer surface **82** is located on a side opposite to the inner surface **81**. The outer surface **82** faces the outer cover **25**.

[0070] The side surface **83** extends substantially in the Z direction between the outer edge of the inner surface **81** and the outer edge of the outer surface **82**. As shown in FIG. **2**, the side surface **83** faces in a direction substantially orthogonal to the central axis Axd. The side surface **83** faces the second inner surface **52**. In the present embodiment, the side surface **83** faces the paint film **22** on the covered region **71**.

[0071] The convex portion **84** protrudes from the side surface **83** toward the exposed region **72**. In the present embodiment, the convex portion **84** has three convex portions **84**A, **84**B, and **84**C. The convex portion **84** is not limited to this example. The three convex portions **84**A, **84**B, and **84**C are spaced apart from each other around the central axis Axd.

[0072] The convex portion **84**A protrudes from the side surface **83** toward the exposed region **72**A. The convex portion **84**B protrudes from the side surface **83** toward the exposed region **72**B. The convex portion **84**C protrudes from the side surface **83** toward the exposed region **72**C.

[0073] Each of the convex portions **84**A, **84**B, and **84**C has a side surface **85**. The side surface **85** faces the exposed region **72**. The side surface **85** of the convex portion **84**A extends substantially parallel to the exposed region **72**A. The side surface **85** of the convex portion **84**B extends substantially parallel to the exposed region **72**B. The side surface **85** of the convex portion **84**C extends substantially parallel to the exposed region **72**C.

[0074] A distance D**3** between the inner cover **23** and the exposed region **72** is smaller than a distance D**4** between the inner cover **23** and the paint film **22**. The distance D**3** is at least one of a distance between the side surface **85** of the convex portion **84**A and the exposed region **72**A, a distance between the side surface **85** of the convex portion **84**B and the exposed region **72**B, and a distance between the side surface **85** of the convex portion **84**C and the exposed region **72**C. The distance D**4** is a distance between the side surface **83** of the inner cover **23** and the paint film **22** on the covered region **71**.

[0075] Around the central axis Axd, a length L**1** of the convex portion **84** is smaller than a length L**2** of the exposed region **72**. The length L**1** is at least one of the length of the convex portion **84**A in the X direction, the length of the convex portion **84**B in the Y direction, and the length of the convex portion **84**C in the Y direction. The length L**2** is at least one of the length of the exposed region **72**A in the X direction, the length of the exposed region **72**B in the Y direction, and the length of the exposed region **72**C in the Y direction.

[0076] The convex portion **84** is located outside the concave portion **75** and is spaced apart from the paint film **22** and the exposed region **72**. The convex portion **84** may be partially accommodated in the concave portion **75** or may be partially in contact with the paint film **22** and the exposed region **72** in other embodiments.

[0077] The gasket **24** is made of, for example, synthetic rubber with low helium permeability. The gasket **24** is formed into a frame shape (endless shape) surrounding the central axes Axc and Axd. The gasket **24** includes a thick portion **91** and a thin portion **92**.

[0078] In the projection plane viewed in the Z direction as shown in FIG. **2**, the thick portion **91** is located between the magnetic disk **12** and the paint film **22**. The thick portion **91** extends substantially along the side surface **83** of the inner cover **23**. However, a portion of the thick portion **91** may be bent, for example, to bypass a fastening member **35** inward.

[0079] In the present embodiment, the thick portion **91** has three thick portions **91**A, **91**B, and **91**C. In the projection plane as shown in FIG. **2**, the thick portion **91**A is located between the magnetic disk **12** and the paint film **22** on the covered region **71**A. The thick portion **91**B is located between the magnetic disk **12** and the paint film **22** on the covered region **71**B. The thick portion **91**C is located between the magnetic disk **12** and the paint film **22** on the covered region **71**C. The thick portion **91** is not limited to this example.

[0080] In the projection plane viewed in the Z direction as shown in FIG. **2**, the thin portion **92** is located between the magnetic disk **12** and the exposed region **72**. The thin portion **92** is bent to protrude from the thick portion **91** toward the exposed region **72**. The thin portion **92** extends along the side surface **85** of the convex portion **84**.

[0081] In the present embodiment, the thin portion **92** has three thin portions **92**A, **92**B, and **92**C. The three thick portions **91**A, **91**B, and **91**C and the three thin portions **92**A, **92**B, and **92**C are disposed alternately around the central axis Axd. The thin portion **92**A is located between the two thick portions **91**A and **91**C. The thin portion **92**B is located between the two thick portions **91**A and **91**B. The thin portion **92**C is located between the two thick portions **91**B and **91**C.

[0082] In the projection plane as shown in FIG. **2**, the thin portion **92**A is located between the magnetic disk **12** and the exposed region **72**A, and is bent to protrude from the thick portions **91**A and **91**C toward the exposed region **72**A. In the projection plane, the thin portion **92**B is located between the magnetic disk **12** and the exposed region **72**B and is bent to protrude from the thick portions **91**A and **91**B toward the exposed region **72**B. In the projection plane, the thin portion **92**C is located between the magnetic disk **12** and the exposed region **72**C and is bent to protrude from the thick portions **91**B and **91**C toward the exposed region **72**C. The thin portions **92** are not limited to this example.

[0083] In the projection plane viewed in the Z direction as shown in FIG. **2**, a width W**1** of the gasket **24** between the magnetic disk **12** and the exposed region **72** is narrower than a width W**2** of the gasket **24** between the magnetic disk **12** and the paint film **22**. In other words, the width W**1** of the thin portion **92** is narrower than the width W**2** of the thick portion **91**. The width of the gasket **24** is not limited to this example.

[0084] For example, SFF-8300, which is a form factor for 3.5-inch hard disk drives established by the Form Factor Committee, sets the maximum dimensions of an HDD. That is, the dimensions of the HDD **10** are restricted in the X direction, the Y direction, and the Z direction. Meanwhile, when the diameter of the magnetic disk **12** is increased, the storage capacity of the HDD **10** can be increased.

[0085] When the diameter of the magnetic disk **12** is increased while the dimension of the HDD **10** is limited, the side wall **32** becomes thinner in the vicinity of the ends P**1**, P**2**, and P**3** of the magnetic disk **12**. In general, when the width of the end face **54** becomes smaller as the side wall **32** becomes thinner, the strength of welding between the end face **54** and the outer cover **25** may be reduced.

[0086] The side wall **32** is protected by the paint film **22**. Electrodeposition coating for forming the paint film **22** has lower thickness accuracy than that of, for example, cutting. In order to prevent interference between the inner cover **23** and the paint film **22**, the inner cover **23** is spaced apart from the paint film **22** on the covered region **71**.

[0087] In the present embodiment, the exposed region **72** is located in the vicinity of the ends P**1**, P**2**, and P**3** of the magnetic disk **12**. The exposed region **72** is formed, for example, by cutting, and thus has relatively high dimensional accuracy. For this reason, a distance between the convex portion **84** of the inner cover **23** and the exposed region **72** can be set to be short.

[0088] As described above, the exposed region **72** can be close to the inner cover **23**. For this reason, the end face **54** connected to the exposed region **72** can also be close to the inner cover **23**. In other words, the end face **54** can expand toward the inner cover **23** in the vicinity of the ends P**1**, P**2**, and P**3** of the magnetic disk **12**.

[0089] On the other hand, the covered region **71** is covered with the paint film **22** but is spaced apart from the magnetic disk **12**. For this reason, even when the covered region **71** is brought close to the magnetic disk **12**, the supporting surface **53** can be provided to have a width so that the inner cover **23** and the gasket **24** can be supported. That is, the end face **54** can be expanded toward the inner cover **23** even at a position spaced apart from the ends P**1**, P**2**, and P**3** of the magnetic disk **12**. For this reason, the end face **54** can have a width that stabilizes the strength of welding.

[0090] The inner cover **23** covers the magnetic disk **12** disposed in the internal space Si. For this reason, the size of the inner cover **23** and the size of the magnetic disk **12** can be correlated. As described above, the inner cover **23** can be close to the exposed region **72**. For this reason, the inner cover **23** and the magnetic disk **12** can be expanded toward the exposed region **72**, and the storage capacity of the magnetic disk **12** can be increased.

[0091] The exposed region **72** is not limited to be formed by cutting, and may be formed by other methods. For example, when the paint film **22** is provided on the second inner surface **52** by electrocoating, a mask may cover the exposed region **72**. The exposed region **72** is exposed by removing the mask.

[0092] In the HDD **10** according to the present embodiment described above, the magnetic disk **12** is rotatable around the central axis Axd. The base **21** is provided with the internal space Si in which the magnetic disk **12** is disposed. The base **21** has the first inner surface **51**, the supporting surface **53**, the second inner surface **52**, and the end face **54**. The first inner surface **51** surrounds the magnetic disk **12**. The supporting surface **53** is connected to the end of the first inner surface **51** in the +Z direction. The second inner surface **52** is further away from the central axis Axd than the first inner surface **51** and is connected to the supporting surface **53**. The end face **54** is connected to the end of the second inner surface **52** in the +Z direction. The paint film **22** covers at least a portion of the first inner surface **51**, at least a portion of the supporting surface **53**, and the covered region **71** of the second inner surface **52**. The paint film **22** is spaced apart from the end face **54** and the exposed region **72** of the second inner surface **52**. The exposed region **72** is aligned with the covered region **71** around the central axis Axd. The inner cover **23** is supported by the supporting surface **53**, surrounded by the second inner surface **52**, and covers the internal space Si. The outer cover **25** is coupled to at least a portion of the base **21** and covers the inner cover **23**.

[0093] The paint film **22** protects a portion of the second inner surface **52**, but may vary in thickness. On the other hand, the exposed region **72** of the second inner surface **52** is not covered with the paint film **22** and is exposed. The exposed region **72** can be formed, for example, by removing the paint film **22** by cutting. For this reason, a distance between the exposed region **72** and the inner cover **23** can be set more accurately and smaller than a distance between the paint film **22** and the inner cover **23**. Thus, in the HDD **10**, the inner cover **23** can be expanded to become close to the exposed region **72**. On the other hand, the exposed region **72** can become close to the inner cover **23**. Thus, in the HDD **10**, the end face **54** connected to the exposed region **72** and the covered region **71** can be expanded toward the inner cover **23**. The outer cover **25** is generally coupled to the base **21** at or near the end face **54**. For this reason, the strength of coupling between the base **21** and the outer cover **25** can be improved by the expansion of the end face **54**. Furthermore, in the HDD **10**, the width of the end face **54** is increased, and thus it is possible to prevent the paint film **22** covering the second inner surface **52** from melting during welding. Furthermore, in the covered region **71**, the paint film **22** covers the second inner surface **52**. For this reason, compared to a case where the paint film **22** is removed over the entire circumference around the central axis Axd, the HDD **10** can reduce work for providing the exposed region **72** and suppress a reduction in the area of the covered region **71** protected by the paint film **22**.

[0094] The exposed region **72** is closer to the magnetic disk **12** than the covered region **71**. As described above, the inner cover **23** can be expanded to become close to the exposed region **72**. Thus, in the HDD **10**, the diameter and storage capacity of the magnetic disk **12** covered by the inner cover **23** can be increased. Furthermore, the covered region **71** is spaced apart from the magnetic disk **12**, and its position can be flexibly set.

[0095] The exposed region **72** has the exposed regions **72**A, **72**B, and **72**C spaced apart from each other around the central axis Axd. The base **21** extends in the +Y direction orthogonal to the central axis Axd. In the projection plane viewed in the +Z direction, the exposed region **72**A faces toward the end P**1** of the magnetic disk **12** in the +Y direction. The exposed region **72**B faces toward the end P**2** of the magnetic disk **12** in the +X direction orthogonal to the +Z direction and the +Y direction. The exposed region **72C** faces the end P**3** of the magnetic disk **12** in the −X direction which is opposite to the +X direction.

[0096] In general, the end P**1** of the magnetic disk **12** in the +Y direction, the end P**2** of the magnetic disk **12** in the +X direction, and the end P**3** of the magnetic disk **12** in the −X direction are closer to the second inner surface **52** than other parts of the magnetic disk **12**. For this reason, although the width of the supporting surface **53** is narrow in the vicinity of each of the ends P**1**, P**2**, and P**3** of the magnetic disk **12** in the +Y direction, the +X direction, and the −X direction, the exposed regions **72**A, **72**B, and **72**C are provided. A distance between each of the exposed regions **72**A, **72**B, and **72**C and the inner cover **23** can be set to be short. Thus, as described above, in the HDD **10**, the diameter of the magnetic disk **12** can be increased, and the strength of coupling between the end face **54** and the outer cover **25** can be improved. Furthermore, the HDD **10** can reduce work for providing the exposed region **72** and suppress a reduction in the area of the covered region **71** protected by the paint film **22**.

[0097] The base **21** further has the second outer surface **57** on a side opposite to the second inner surface **52**. The distance D**1** between the exposed region **72** and the second outer surface **57** is smaller than the distance D**2** between the covered region **71** and the second outer surface **57**.

[0098] For example, the exposed region **72** that is exposed without being covered with the paint film **22** can be formed by cutting the paint film **22** and the covered region **71**. By forming the exposed region **72**, as described above, it is possible to increase the diameter of the magnetic disk **12** and improve the strength of coupling between the end face **54** and the outer cover **25** in the HDD **10**. Further, the width of the end face **54** may be set to be wide between the covered region **71** and the second outer surface **57**. For this reason, in the HDD **10**, it is possible to improve the strength of coupling between the end face **54** and the outer cover **25**.

[0099] The distance D**3** between the inner cover **23** and the exposed region **72** is smaller than the distance D**4** between the inner cover **23** and the paint film **22**. Thereby, as described above, in the HDD **10**, it is possible to increase the diameter of the magnetic disk **12** and to improve the strength of coupling between the end face **54** and the outer cover **25**.

[0100] The inner cover **23** has the side surface **83** facing the paint film **22** and the convex portion **84** protruding from the side surface **83** toward the exposed region **72**. Thereby, in the HDD **10**, a distance between the inner cover **23** and the exposed region **72** can be set to be short, and the side surface **83** and the paint film **22** can be prevented from interfering with each other.

[0101] Around the central axis Axd, the length L**1** of the convex portion **84** is smaller than the length L**2** of the exposed region **72**. Thereby, the convex portion **84** can be prevented from interfering with the paint film **22** in the vicinity of the exposed region **72**.

[0102] The gasket **24** is interposed between the supporting surface **53** and the inner cover **23**. The gasket **24** includes the thick portion **91** and the thin portion **92**. The thick portion **91** is located between the magnetic disk **12** and the paint film **22** on the projection plane viewed in the +Z direction. The thin portion **92** is located between the magnetic disk **12** and each of the exposed regions **72** in the projection plane viewed in the +Z direction, and is bent to protrude from the thick portion **91** toward the exposed regions **72**.

[0103] The thin portion **92** is bent to be spaced apart from the edge of the supporting surface **53** connected to the first inner surface **51** at positions corresponding to the exposed region **72** and the convex portion **84**. Thereby, the gasket **24** can prevent the thin portion **92** from falling off from the supporting surface **53** and can further reliably seal the internal space Si.

[0104] The gasket **24** is interposed between the supporting surface **53** and the inner cover **23**. In the projection plane viewed in the +Z direction, the width W**1** of the gasket **24** between the magnetic disk **12** and the exposed region **72** is narrower than the width W**2** of the gasket **24** between the magnetic disk **12** and the paint film **22**.

[0105] Since the exposed region **72** is closer to the magnetic disk **12** than the covered region **71**, the width of the supporting surface **53** is narrower in the vicinity of the exposed region **72**. Since the width W**1** of the gasket **24** is narrow in the vicinity of the exposed region **72**, the gasket **24** can be prevented from falling off from the supporting surface **53** and can more reliably seal the internal space Si.

[0106] The outer cover **25** is coupled to the end face **54**. As described above, in the HDD **10**, the end face **54** can be expanded toward the inner cover **23**, and the strength of coupling between the end face **54** and the outer cover **25** can be improved.

[0107] The base **21** further has the inclined surface **55**. The inclined surface **55** extends obliquely between the second inner surface **52** and the end face **54** with respect to the second inner surface **52** and the end face **54**. The length of the inclined surface **55** between the exposed region **72** and the end face **54** is smaller than the length of the inclined surface **55** between the covered region **71** and the end face **54**.

[0108] The inclined surface **55** is set to be short, and thus it is possible to prevent the width of the end face **54** from decreasing in the vicinity of the exposed region **72**. For this reason, in the HDD **10**, the strength of coupling between the end face **54** and the outer cover **25** can be improved.

[0109] The diameter of the magnetic disk **12** is 96 mm or more. In general, when the diameter of the magnetic disk **12** is set to be large, a distance between the second inner surface **52** and the magnetic disk **12** decreases. However, as described above, in the HDD **10**, the diameter of the magnetic disk **12** can be increased, and the strength of coupling between the end face **54** and the outer cover **25** can be improved.

[0110] The exposed region **72** is spaced apart from the supporting surface **53** and faces the inner cover **23**. Thereby, in the HDD **10**, it is possible to prevent the paint film **22** covering the supporting surface **53** from being removed during cutting for forming the exposed region **72** and to prevent the area of the supporting surface **53** protected by the paint film **22** from being reduced.

[0111] The inner cover **23** is further away from the supporting surface **53** than the end of the exposed region **72** in the −Z direction opposite to the +Z direction. Thereby, in the HDD **10**, it is possible to prevent the inner cover **23** and the paint film **22** provided between the supporting surface **53** and the exposed region **72** from interfering with each other.

[0112] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

What is claimed is:

1. A disk device comprising:

a magnetic disk that is rotatable around a rotation axis;

a base that is provided with an internal space in which the magnetic disk is disposed, and has a first inner surface surrounding the magnetic disk, a supporting surface connected to an end of the first inner surface in a first direction parallel to the rotation axis, a second inner surface further away from the rotation axis than the first inner surface and connected to the supporting surface, and an end face connected to an end of the second inner surface in the first direction;

a film that covers at least a portion of the first inner surface, at least a portion of the supporting surface, and a covered region of the second inner surface, and is spaced apart from the end face and an exposed region of the second inner surface;

an inner cover that is supported by the supporting surface, surrounded by the second inner surface, and covering the internal space; and

an outer cover that is coupled to at least a portion of the base and covering the inner cover.

2. The disk device according to claim **1**, wherein the exposed region is closer to the magnetic disk than the covered region.

3. The disk device according to claim **2**, wherein

the exposed region includes a first exposed region, a second exposed region, and a third exposed region which are spaced apart from each other around the rotation axis,

the base extends in a second direction orthogonal to the rotation axis, and

when viewed in the first direction, the first exposed region faces an end of the magnetic disk in the second direction, and the second exposed region faces an end of the magnetic disk in a third direction orthogonal to the first direction and the second direction, and the third exposed region faces an end of the magnetic disk in a fourth direction opposite to the third direction.

4. The disk device according to claim **1**, wherein

the base further has an outer surface on a side opposite to the second inner surface, and a distance between the exposed region and the outer surface is smaller than a distance between the covered region and the outer surface.

5. The disk device according to claim **1**, wherein a distance between the inner cover and the exposed region is smaller than a distance between the inner cover and the film.

6. The disk device according to claim **5**, wherein the inner cover has a side surface facing the film, and a convex portion protruding from the side surface toward the exposed region in a second direction orthogonal to the first direction.

7. The disk device according to claim **6**, wherein a length of the convex portion in a third direction orthogonal to the first and second directions is smaller than a length of the exposed region in the third direction.

8. The disk device according to claim **6**, further comprising a gasket that is interposed between the supporting surface and the inner cover, and includes a first portion located between the magnetic disk and the covered region when viewed in the first direction, and a second portion that is located between the magnetic disk and the exposed region when viewed in the first direction and is bent to protrude from the first portion toward the exposed region.

9. The disk device according to claim **1**, further comprising a gasket that is interposed between the supporting surface and the inner cover, wherein

when viewed in the first direction, a width of the gasket that is between the magnetic disk and the exposed region is narrower than a width of the gasket that is between the magnetic disk and the covered region.

10. The disk device according to claim **1**, wherein the outer cover is coupled to the end face.

11. The disk device according to claim **10**, wherein

the base further includes an inclined surface that extends obliquely between the second inner surface and the end face with respect to the second inner surface and the end face, and

a length of the inclined surface between the exposed region and the end face is smaller than a length of the inclined surface between the covered region and the end face.

12. The disk device according to claim **1**, wherein a diameter of the magnetic disk is 96 mm or more.

13. The disk device according to claim **1**, wherein the exposed region is spaced apart from the supporting surface and faces the inner cover.

14. The disk device according to claim **13**, wherein the inner cover is further away from the supporting surface than an end of the exposed region in a direction that is opposite to the first direction.

15. A disk device comprising:

a magnetic disk;

a base having an internal space in which the magnetic disk is disposed, the base including an inner surface surrounding the magnetic disk and extending in a first direction parallel to a rotation axis of the magnetic disk, a supporting surface connected to an end of the first inner surface in the first direction and extending in a second direction orthogonal to the first direction, a second inner surface further away from the rotation axis than the first inner surface, connected to the supporting surface, and extending in the first direction, and an end face connected to an end of the second inner surface in the first direction and extending in the second direction;

a film that covers the first inner surface, the supporting surface, and first, second, and third parts of the second inner surface, each of which extends partially around the rotation axis, and that does not cover the end face, a fourth part of the second inner surface that is between the first and second parts of the second inner surface, a fifth part of the second inner surface that is between the first and third parts of the second inner surface, and a sixth part of the second inner surface that is between the second and third parts of the second inner surface;

an inner cover that is supported by the supporting surface, surrounded by the second inner surface, and covering the internal space of the base; and

an outer cover that is coupled to at least a portion of the base and covering the inner cover.

16. The disk device according to claim **15**, wherein the inner cover has convex portions, each of which faces the fourth, fifth, and sixth parts of the second inner surface, in a direction that is radially outward with respect to the rotation axis.

17. The disk device according to claim **15**, further comprising a gasket that is interposed between the supporting surface and the inner cover.

18. The disk device according to claim **17**, wherein the gasket is provided contiguously around the rotation axis and has thick portions that are aligned with the first, second, and third parts of the second inner surface in the first direction and thin portions that are aligned with the fourth, fifth, and sixth parts of the second inner surface in the first direction.

19. The disk device according to claim **15**, wherein the outer cover is coupled to the end face.

20. The disk device according to claim **19**, wherein

the base further includes an inclined surface that extends obliquely between the second inner surface and the end face with respect to the second inner surface and the end face.

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